Incorporating Tools and Memory in Agents

Agenda

- Tools in AI Agents
- Memory in AI Agents
- Model Context Protocol

Let’s begin the discussion by answering a few questions.

Agentic AI Quiz

In an agent-based system, tools are introduced mainly to enable the agent to go beyond text generation. Which of the following best explains why tools are useful?

A. They make the agent respond faster by skipping reasoning steps

B. They allow the agent to interact with external systems such as databases or APIs

C. They remove the need for prompts and instructions

D. They restrict the agent's behavior to predefined responses

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Tools

Tools are external functions or systems that an agent can use to take actions beyond text generation.

Why do agents need tools?

- To interact with real data (databases, files, APIs)
- To perform deterministic operations (calculations, queries)
- To reduce hallucinations by fetching factual information

Examples of tools used by agents

- SQL tool to query a database
- API tool to fetch live data (orders, weather, prices)
- Calculator or code execution tool

Agentic AI Quiz

What is the primary benefit of adding memory to agents?

- A. Decreasing computational cost
- B. Eliminating the need for data verification
- C. Providing personalization and context continuity
- D. Making the agent slower to respond

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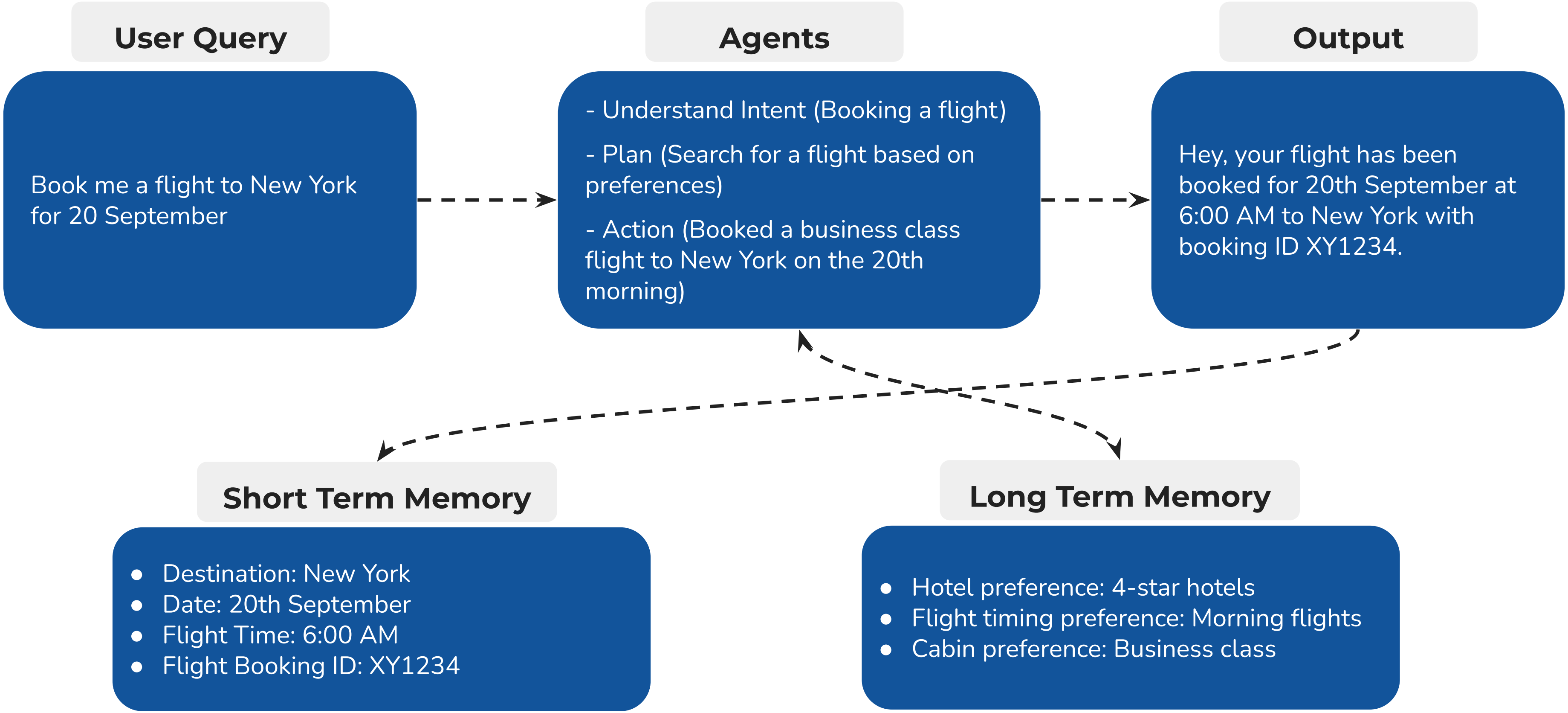
Memory in Agents

Memory enables an agent to retain and reuse information across interactions, helping it maintain context, continuity, and personalization instead of treating every query as a standalone request.

Short-term memory	Long-term memory
Used within a single conversation to track recent user inputs, instructions, and intermediate reasoning	Used across multiple conversations to retain important user information, preferences, or historical interactions.
Ensures coherent, context-aware responses within the ongoing interaction	Enables personalization, consistency, and improved user experience over time
Stored in the conversation context or prompt window	Stored in external systems such as databases, vector stores, or memory services

Memory in Agents



Memory in Agents

User Query (Follow up)

Book me a Hotel near Times Square

Agents

- Understand Intent (Booking a hotel)
- Plan (Search for a 4 Star hotel near Square Times for 20th September)
- Action (Booked a Hotel)

Output

Hey, your ABC Hotel has been booked for 20th September with booking ID AB1234.

Short Term Memory

- Destination: New York
- Date: 20th September
- Flight Time: 6:00 AM
- Flight Booking ID: XY1234
- ABC Hotel
- Hotel Booking ID - AB1234

Long Term Memory

- Hotel preference: 4-star hotels
- Flight timing preference: Morning flights
- Cabin preference: Business class

Agentic AI Quiz

What is the primary purpose of the Model Context Protocol (MCP) in Agentic AI systems?

A. To improve GPU performance during model training

B. To standardize how agents connect with external tools and data sources

C. To replace traditional machine learning models with LLMs

D. To ensure that memory in LLMs never expires

Agentic AI Quiz

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Model Context Protocol (MCP)

Agents often need to work with many external systems (databases, APIs, files).

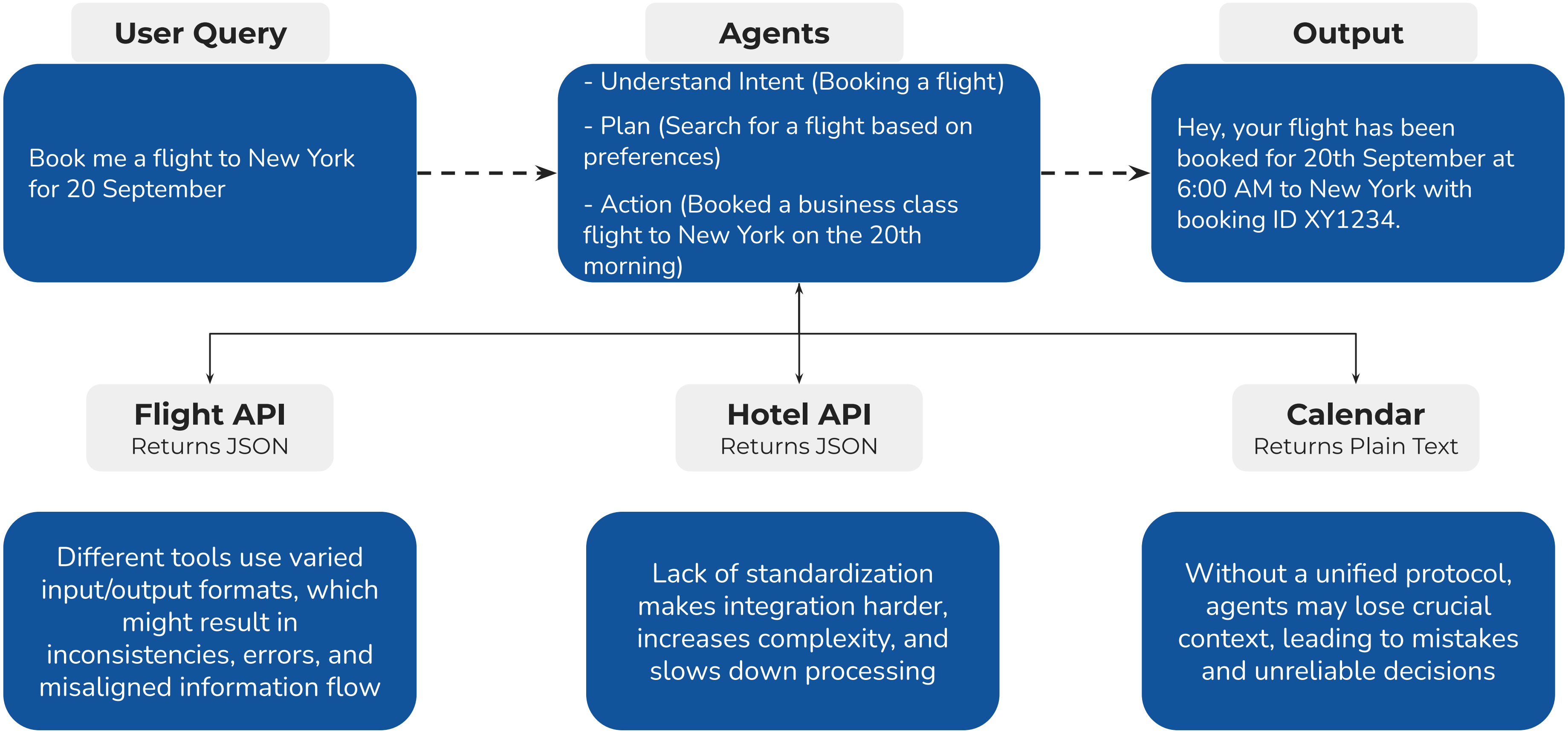
Without a standard, each integration becomes tightly coupled and hard to scale.

MCP solves this by defining a common protocol for how agents discover and use external capabilities.

How MCP works	Why MCP matters
External systems expose their capabilities through an MCP Server	Decouples agent logic from tool implementation
The agent connects as an MCP Client	Makes integrations reusable and maintainable
The agent discovers available tools and invokes them in a controlled, structured way	Enables consistent, enterprise-ready agent architectures

Model Context Protocol (MCP)

Great Learning
POWER AHEAD



Model Context Protocol (MCP)

An **open standard** that solves integration and context-loss issues by **enabling AI agents to reliably discover, access, and use tools** in a **consistent, unified way** at runtime.

User Query

Book me a flight to New York for 20 September

Agents

- Understand Intent (Booking a flight)
- Plan (Search for a flight based on preferences)
- Action (Booked a business class flight to New York on the 20th morning)

Output

Hey, your flight has been booked for 20th September at 6:00 AM to New York with booking ID XY1234.

MCP

Flight API

Hotel API

Calendar

Great Learning

Power Ahead!